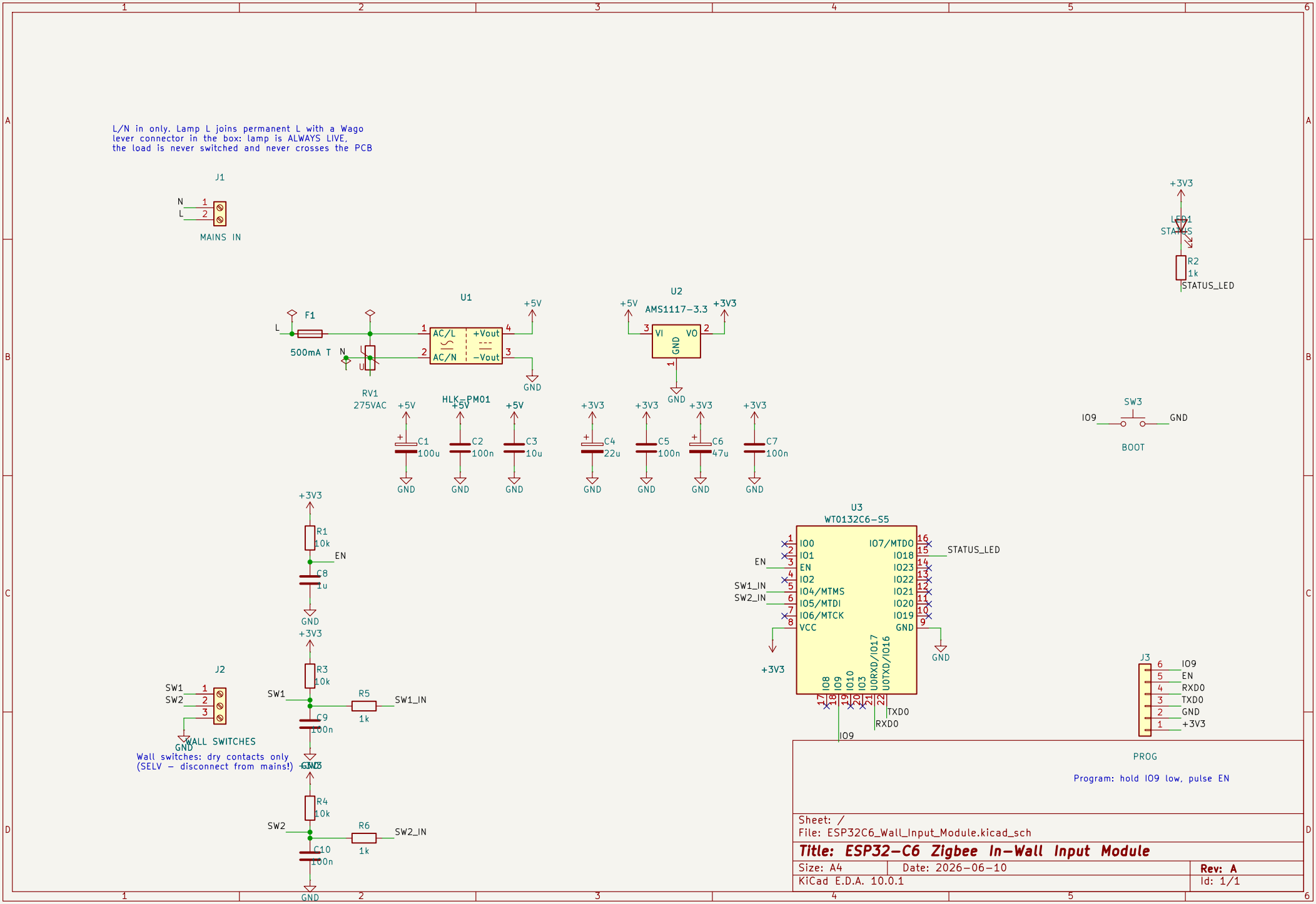
L/N in only. Lamp L joins permanent L with a Wago
lever connector in the box: lamp is ALWAYS LIVE,
the load is never switched and never crosses the PCB
J1
N
L
MAINS IN
F1
500mA T
U1
AC/L
AC/N
+Vout
-Vout
+5V
GND
RV1
275VAC
HLK-PM01
C1
100u
C2
100n
C3
10u
U2
AMS1117-3.3
VI
VO
GND
+3V3
C4
22u
C5
100n
C6
47u
C7
100n
+3V3
LED1
STATUS
R2
1k
STATUS_LED
SW3
IO9
GND
BOOT
U3
WT0132C6-S5
IO0
IO1
EN
IO2
IO4/MTMS
IO5/MTDI
IO6/MTCK
VCC
IO7/MTDO
IO18
IO23
IO22
IO21
IO20
IO19
GND
IO8
IO9
IO10
IO3
U0RXD/IO17
U0TXD/IO16
SW1_IN
SW2_IN
STATUS_LED
TXD0
RXD0
J3
IO9
EN
RXD0
TXD0
GND
+3V3
PROG
Program: hold IO9 low, pulse EN
J2
SW1
SW2
WALL SWITCHES
Wall switches: dry contacts only
(SELV – disconnect from mains!)
R1
10k
C8
1u
R3
10k
R5
1k
C9
100n
R4
10k
R6
1k
C10
100n
SW1_IN
SW2_IN
Sheet: /
File: ESP32C6_Wall_Input_Module.kicad_sch
Title: ESP32-C6 Zigbee In-Wall Input Module
Size: A4
Date: 2026-06-10
Rev: A
KiCad E.D.A. 10.0.1
Id: 1/1